

# A Quantitative Security Model for Trapdoor Computing

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## Abstract

Trapdoor computing lets an untrusted machine evaluate opaque bit strings through an opaque lookup table while only a trusted machine encodes and decodes. Its central claim is that confidentiality is *measured*, an explicit leakage quantity a designer can trade, rather than *negligible* in a security parameter. That claim has so far rested on an informal model: the constructions come with leakage *measures*, but not with a security *definition* tying an adversary’s power to what is provably guaranteed. This paper supplies the missing model. We formalize the trusted/untrusted adversary and its view, and define the confidentiality of a cipher map in the language of quantitative information flow, as a tuple of bounded, composable leakage quantities rather than a negligible advantage. We show that the framework’s three leakage measures, the marginal entropy ratio, the compositional joint-recovery rate, and the active orbit-closure bound, are three aspects of one adversary with three kinds of access, and that the resulting security tuple composes under cipher-map composition. We close by relating the definition to simulation-based and game-based cryptographic security, arguing that quantitative information flow, not indistinguishability, is the right model for the measurable-leakage regime trapdoor computing occupies.

## 1 Introduction

A security definition does two things: it says what an adversary may do, and it says what is guaranteed against an adversary who does it. Trapdoor computing has a clear adversary, the untrusted machine, and a clear collection of guarantees, the leakage bounds its constructions satisfy, but it has not had a definition that joins the two. The cipher-maps abstraction states informally that the untrusted machine “cannot decode, cannot distinguish a real query from filler, cannot determine the domain, cannot be sure a result is correct” [15]; the confidentiality theory gives an entropy-ratio bound on a single token and a recovery-rate bound on a chain [16]; the type theory gives an orbit-closure bound on an active adversary [14]. These are measures in search of a model.

The gap is not cosmetic. A reviewer trained on simulation-based security asks, reasonably, for a definition: an adversary, a security experiment, and a statement of what the experiment guarantees. Absent one, “measured, not negligible” reads as a slogan rather than a claim. This paper makes it a claim. We give the formal trusted/untrusted model and the adversary’s view (Section 2); define a cipher map’s confidentiality as a tuple of bounded leakage quantities in the sense of quantitative information flow [13, 1] (Section 3); show the framework’s three leakage measures are one adversary at three access levels (Section 4); prove the security tuple composes (Section 5); and defend the choice of a quantitative rather than a simulation-based or game-based definition (Section 6).

The contribution is the definition itself, and the demonstration that the framework’s existing, separately-proven bounds are instances of it. We prove no new leakage bound here; we give the model the bounds were always bounds in.

## 2 The Trusted/Untrusted Model

Fix a latent function  $f : X \rightarrow Y$  known only to a trusted party.

**Definition 1** (Cipher map; from [15]). A *cipher map* for  $f$  is a tuple  $(\hat{f}, \text{enc}, \text{dec}, s)$  with  $\hat{f} : \{0, 1\}^n \rightarrow \{0, 1\}^n$  a total function,  $\text{enc} : X \times \{0, \dots, K(x) - 1\} \rightarrow \{0, 1\}^n$ ,  $\text{dec} : \{0, 1\}^n \rightarrow Y \cup \{\perp\}$ , and  $s$  a secret from which all three derive. The *trusted machine*  $T$  holds  $(\text{enc}, \text{dec}, s)$ ; the *untrusted machine*  $U$  holds  $\hat{f}$ .

The adversary is  $U$ . We make its power and its view explicit.

**Definition 2** (Adversary view). In a run with in-domain inputs  $x_1, \dots, x_N$  drawn i.i.d. from a query distribution  $D$  on  $X$ , the untrusted machine’s view is

$$\text{view}_U = (\hat{f}, (c_i)_{i=1}^N, (\hat{f}(c_i))_{i=1}^N, F),$$

where  $c_i = \text{enc}(x_i, k_i)$  is the  $i$ -th query token, and  $F$  is the set of cipher maps  $U$  holds and may apply to any token to produce and observe further tokens.  $U$  is unbounded for the purpose of the leakage analysis: we measure how much  $\text{view}_U$  reveals, not whether a bounded  $U$  can extract it.

Two modeling choices deserve comment. First,  $\hat{f}$  is *public to  $U$* : the untrusted machine holds the whole table, so the model grants the adversary the construction itself and asks what the construction still hides. Second, totality closes one adversarial channel: because every token, honest or not, produces an answer,  $U$  gains nothing from an out-of-band error or exception signal, as there is none to probe. This does not make  $U$  passive. The operations  $F$  in its view already let it form and probe self-generated tokens, the active capability that clause (A) measures, so the model grants the active power; totality removes only the error channel. (The marginal bound below additionally assumes the membership set is unstructured; an adversary exploiting algebraic structure in the key universe, for example  $S = \{x : x \equiv 0 \bmod 7\}$ , is out of scope and is handled separately [16].)

The latent quantities  $U$  would learn, and that the model must bound, are the inputs  $X = (x_i)$ , the function  $f$ , the distribution  $D$ , and quantities derived from them: the domain, the equality structure of the queries, and the joint distribution of correlated values across several cipher maps.

## 3 Confidentiality as Bounded Leakage

Quantitative information flow measures the leakage of a secret  $S$  to a view  $V$  by how much  $V$  reduces uncertainty about  $S$ , for instance the Shannon leakage  $I(S; V)$ , a min-entropy (Bayes-vulnerability) leakage, or maximal leakage [9]. Security in this language is not the absence of leakage but a *bound* on it. We define a cipher map’s confidentiality as a tuple of such bounds, one per kind of access in Definition 2.

**Definition 3** (Confidentiality of a cipher map). Let  $(\hat{f}, \text{enc}, \text{dec}, s)$  be a cipher map with representation uniformity  $\delta = \text{TV}(Q, U_{\text{im}})$  on its populated support, where  $Q$  is the induced

token distribution and  $U_{\text{im}}$  is uniform on  $\text{im}(\text{enc})$ , and let  $H^* = \log_2 |\text{im}(\text{enc})|$ . The cipher map is  $(\delta, R, F)$ -*confidential* against the adversary of Definition 2 if all three of the following hold, where  $\delta$  is the marginal parameter of (M),  $R$  the recovery-rate function of (C), and  $F$  the operation set of (A); the cipher map’s confidentiality is the triple, not any one of them.

- (M) Marginal. The observed single-token distribution  $Q$  is  $\delta$ -close to uniform on the support, so its entropy ratio is high,

$$e = \frac{H(Q)}{H^*} \geq 1 - \delta - \frac{h_2(\delta)}{H^*},$$

the Fannes–Audenaert lower bound [7, 3], linear in  $\delta$ ; and a single-guess membership-inference adversary observing one decoded output distinguishes a real query from filler with advantage at most  $\delta/2$  (Le Cam’s two-point method). Small  $\delta$  bounds both the stream’s non-uniformity,  $1 - e = D_{\text{KL}}(Q \| U_{\text{im}})/H^*$ , and the single-guess advantage.

- (C) Compositional. For two cipher maps  $\hat{f}_1, \hat{f}_2$  for  $f_1 : X \rightarrow Y_1$ ,  $f_2 : X \rightarrow Y_2$  observed on shared tokens, the latent joint distribution on  $Y_1 \times Y_2$  leaks: it is recoverable to total-variation accuracy  $\xi$  once  $N = R(\xi) = \Theta(|Y_1| |Y_2| / \xi^2)$  shared observations are taken, and no construction parameter lowers this rate. The maps preserve the latent dependence, with mutual information preserved exactly,  $I(\hat{f}_1(C); \hat{f}_2(C)) = I(f_1(X); f_2(X))$ , when the outputs are canonical ( $K_{\text{out}} = 1$ ) and  $\geq$  in general. This is the Dalenius (correlated-secrets) leakage, here with an explicit rate.

- (A) Active. For an adversary applying the operations  $F$  to a token  $c$ , with orbit-resolved observation  $\mathcal{V}_F(c)$  supported on the reachable set  $\text{orbit}_F(c)$ , the residual entropy of the latent value  $X$  is

$$H(X | \mathcal{V}_F(c)) \geq H(X) - \log_2 |\text{orbit}_F(c)|,$$

equivalently  $\text{conf} \geq 1 - |\text{orbit}_F(c)|/|X|$  in set form, the denominator being the latent space  $|X|$ , not  $2^n$ .

The definition is parameterized two ways: by the *construction* (through  $\delta$  and the acceptance partition that shapes the orbit and the recovery constants) and by the *adversary’s access* (through the number of shared observations  $N$  and the operation set  $F$ ). A cipher map is not “secure” or “insecure” absolutely; it is secure to these explicit, computable degrees, and the degrees are exactly the quantities a designer trades against space, bandwidth, and accuracy. This is the formal content of “measured, not negligible.”

*Remark 1* (Why a tuple, not a single number). The three clauses cannot be collapsed into one bound, because they are not comparable quantities. (M) is a normalized leakage of a single value, governed by a construction parameter. (C) is a *rate* at which a joint distribution becomes estimable, governed by the latent cardinalities and unaffected by any construction parameter. (A) is an unnormalized residual entropy, governed by the exposed operations. A security model for trapdoor computing must carry all three because a cipher map can be strong on one and weak on another: near-perfect marginal confidentiality ( $\delta \approx 0$ ) coexists with full compositional recovery given enough shared observations. The irreducibility of the scales is a feature of the model, not a defect of a construction. In the language of generalized gain functions [2], the three clauses are three gain functions on the one adversary view of Definition 2, each scoring a different operational goal, guessing one value, recovering a joint distribution, or narrowing a value by operating, and the model’s content is that no single gain function dominates the three.

Table 1: The three clauses are one adversary (Definition 2) at three access levels.

| Clause            | Access used            | Governed by              | Bounded by $\delta$ ? |
|-------------------|------------------------|--------------------------|-----------------------|
| (M) marginal      | one token, passive     | $\delta$ (entropy ratio) | yes                   |
| (C) compositional | shared tokens, passive | $ Y_1  Y_2 , N$          | no                    |
| (A) active        | operations $F$         | $ \text{orbit}_F(c) $    | no                    |

## 4 Three Scales, One Adversary

The three clauses of Definition 3 are often read as three adversaries; they are one adversary at three levels of access (Table 1). Clause (M) is the adversary of Definition 2 restricted to a single token and passive. Clause (C) is the same adversary, still passive, but observing several cipher maps on shared tokens, the access that exposes joint structure. Clause (A) is the adversary using the last component of its view, the operations  $F$ , to probe actively. Nothing in the three requires different powers; they partition the single view of Definition 2 by which part of it the adversary exploits.

The table makes the model’s sharpest statement visible. Only (M) is bought by a construction parameter. The compositional and active leakages are governed by quantities  $\delta$  does not touch, the latent cardinalities and the number of observations for (C), the reachable orbit for (A). A designer who lowers  $\delta$  to its floor has done everything the marginal scale allows and nothing for the other two; defending those is a matter of system design, limiting shared observations and exposed operations, not of tuning a parameter. The model thus predicts where engineering effort can and cannot help.

## 5 Composition

A security definition is useful for building systems only if it composes: the security of a composite must be computable from the security of its parts. The tuple of Definition 3 does.

**Proposition 1** (Composition of the security tuple). *Let  $\hat{f}_m \circ \dots \circ \hat{f}_1$  be a chain of cipher maps, stage  $i$  with correctness  $\eta_i$  and marginal parameter  $\delta_i$ , forming a cipher map for the composite latent function. Three of the four components of its security tuple are bounded by the parts:*

1. (correctness)  $\eta_{\text{total}} \leq 1 - \prod_{i=1}^m (1 - \eta_i)$ ;
2. (marginal) *each token the untrusted machine observes, the query and every intermediate output, is a single token whose marginal leakage clause (M) bounds at that token’s own  $\delta_i$ , so composition does not amplify the per-token marginal leakage;*
3. (active) *for a typed chain the reachable orbit is bounded additively,  $|\text{orbit}_F| \leq \sum_i N_i$  with  $N_{i+1} = N_i^{a_i}$  for per-step arity  $a_i$  [14], so clause (A) holds at design time, independent of the cipher-space sizes.*

*Proof.* (1) The chain is correct whenever every stage is, since a stage given a correct intermediate value answers correctly up to its own error. Writing  $C_i$  for the event that stage  $i$  is correct given correct input,  $\bigcap_i C_i$  implies the chain is correct, so  $\Pr[\text{chain correct}] \geq \prod_i \Pr[C_i \mid C_1, \dots, C_{i-1}] \geq \prod_i (1 - \eta_i)$ , the last step because, given correct upstream, stage  $i$  errs with probability at most  $\eta_i$ .

Complementing gives the bound; it is an inequality, not an equality, because a stage error can be undone by a later stage mapping the corrupted token to a correct value [15]. (2) Clause (M) bounds the leakage of a *single* observed token by its representation-uniformity parameter, through the Fannes and single-guess bounds. Every token the chain exposes is one such token, governed by the  $\delta_i$  of the map that produced it; since (M) is a per-token bound, the chain’s marginal leakage at any one observation is that token’s bound, with no amplification. (3) The typed-composition discipline of [14] bounds the reachable set of a depth- $m$  typed chain by the sum of its per-step frontiers  $N_i$ , with the recurrence  $N_{i+1} = N_i^{a_i}$ ; substituting into clause (A) bounds the active residual entropy by a quantity the typed structure fixes, not by  $2^n$ .  $\square$

The proposition deliberately stops at three components. It does *not* settle the compositional component (C), and the gap is the one quantitative information flow calls *collateral leakage* [11, 10]: composing channels can leak jointly more than the sum of their marginal leakages, so whether a cipher-map chain exposes joint structure across its stages that no stage exposes alone is not decided by parts (1)–(3). The negative half *is* known: the chain’s joint-recovery rate is governed by the latent cardinalities it exposes (clause (C)), and no per-stage parameter lowers it, so the compositional component cannot be made to compose by tuning the parts. A full composition law for (C), accounting for collateral leakage, is the principal formal work this model leaves open; until it exists, the compositional component must be controlled directly, by limiting the shared-token observations an adversary can make. The other three components compose through quantities the parts already carry, which is what lets a designer estimate them from a parts list and lets the definition support construction, not only analysis.

## 6 Relation to Cryptographic Security

The natural objection to Definition 3 is that it is not a cryptographic security definition. It is not, and the difference is the point.

**Against simulation-based security.** Searchable encryption is proved secure by exhibiting a simulator that reproduces the adversary’s view from a *leakage function* [5]: security means the real view is indistinguishable from a simulation depending only on what the leakage function reveals, and the leakage-abuse literature then studies what that documented function lets an attacker recover [12, 4]. Oblivious RAM takes a different route, proving access-pattern indistinguishability rather than exhibiting a leakage function [8]. Trapdoor computing’s leakage is not a function to be fed to a simulator; it is a family of *bounds*, and the constructions do not claim the residual is simulatable, only that it is small in the measured sense. One could define a leakage function and seek a simulator, but the resulting statement would say less than Definition 3: a simulator with a large leakage function certifies indistinguishability while hiding how much leaks, whereas the quantitative bound says exactly how much.

**Against game-based security.** Indistinguishability definitions promise a *negligible* advantage in a security parameter. Trapdoor computing makes no such promise, and a construction forced to claim one would either fail, because its compositional leakage is not negligible, or mislead. The honest statement is the quantitative one: the advantage is a measured quantity, often not negligible, and useful precisely because it is known.

**Against prior quantitative treatments of encrypted search.** Quantitative information flow has been brought to encrypted search before. The q-leakage analysis of [6] replaces the leakage-function-and-simulator paradigm with a measured leakage quantity, exactly the move this paper makes for trapdoor computing, and the broader QIF programme [13, 1, 2] established security-as-a-bound, the irreducibility of leakage measures across operational scenarios, and the correlated-secrets (Dalenius) leakage that clause (C) instantiates. The contribution here is therefore not the quantitative stance but its specialization to the opaque-table adversary: the three-clause tuple tied to one adversary’s three access levels, and in particular the active orbit-closure clause (A), which passive access-pattern leakage models do not capture, since their adversary observes rather than *operates*.

**Why quantitative information flow is the right model.** The leakage trapdoor computing incurs is operationally meaningful (its bounds are the quantities a designer trades), composable (Proposition 1), and honest (it does not assert a negligibility it cannot support). These are exactly the virtues quantitative information flow was built to express. The cost is real: the guarantee is weaker than indistinguishability, and for an application that can afford the strong tools, the strong tools are correct. For the regime between plaintext and oblivious RAM, where leakage is unavoidable but bounded and tradeable, the quantitative definition is not a weaker substitute for the cryptographic one; it is the right definition.

## 7 Conclusion

We have given trapdoor computing the security model its measurable-leakage claim required: a formal adversary and view, a definition of confidentiality as a tuple of bounded leakage quantities in the sense of quantitative information flow, a demonstration that the framework’s three leakage measures are one adversary at three access levels, and a proof that the tuple composes. The model proves no new bound; it gives the bounds the framework already had a definition to be bounds in, and in doing so answers the question a foundations reviewer asks first: what does it mean, here, to be secure? It means leaking a measured, composable, and computable amount, and no more.

**Provenance.** The trapdoor framework was begun in 2017 and developed through 2020, then recovered and extended in 2024; this model formalizes the security stance implicit in it throughout.

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